4-bit Array Multiplier from RTL to GDSII on SkyWater SKY130 with Open-Source Tools

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Digital VLSI Design — RTL-to-GDSII Assignment
Repository: https://github.com/SudhuJ/arrayMultiplier

***Abstract*—We design a 4-bit $\times$ 4-bit unsigned array multiplier in Verilog with registered inputs and output, verify it exhaustively, and take it through synthesis (Yosys), static timing analysis (OpenSTA) and place and route (LibreLane) on the SKY130 high-density standard-cell library. All timing is evaluated at the typical corner ($25\,^\circ\mathrm{C}$, $1.80\,\mathrm{V}$). The RTL and the gate-level netlist both pass all 256 test vectors. Synthesis produces 61 cells with an area of $757.0\,\mu\mathrm{m}^2$, $52.9\,\%$ of it flip-flops. STA gives $T_{\min} = 2.79\,\mathrm{ns}$ ($F_{\max} = 358.4\,\mathrm{MHz}$). The final layout, at a $3.5\,\mathrm{ns}$ clock, has 0 DRC errors, 0 LVS errors and $0.077\,\mathrm{ns}$ of setup slack, giving a post-route $F_{\max}$ of $292.2\,\mathrm{MHz}$.**

I. Introduction

An array multiplier computes a product with a regular grid of AND gates and full adders, which makes it a good first design to take through a complete physical-design flow: its structure is easy to check by hand, yet it exercises every stage of the flow. Fig. 1 shows the flow we followed and the tool used at each stage. Every stage is checked before moving on:

- the RTL is verified exhaustively in simulation (Part A);
- the synthesised netlist is simulated again with the same testbench (Part B);
- static timing analysis gives the maximum clock frequency (Part C);
- the layout is signed off with timing, DRC and LVS checks (Part D).

II. Design

A. Array structure

For every bit b_i of B there is a partial product $pp_i[j] = a[j]\cdot b[i]$, shifted left by i. Row 0 consists only of AND gates: it gives $p[0] = pp_0[0]$ and passes $r_0 = \{0, pp_0[3{:}1]\}$ down. Rows 1–3 are 4-bit ripple-carry adders. Full adder j of row i adds $pp_i[j]$, $r_{i-1}[j]$ and the carry $c_i[j]$, with $c_i[0] = 0$. The lowest sum bit of each row is a finished product bit, $p[i] = s_i[0]$, and $r_i = \{c_i[4], s_i[3{:}1]\}$ goes to the next row. Row 3 also gives $p[7{:}4] = \{c_3[4], s_3[3{:}1]\}$. The array therefore uses $N^2 = 16$ AND gates and $N(N-1) = 12$ full adders (Fig. 3).

B. RTL

`full_adder` is written with two `assign` statements, $sum = a \oplus b \oplus c_{\mathrm{in}}$ and $c_{\mathrm{out}} = ab + a\,c_{\mathrm{in}} + b\,c_{\mathrm{in}}$. `mult_array` registers a and b into a_r and b_r on every rising edge, builds the three adder rows with `generate` loops (12 `full_adder` instances, no $\star$ operator) and registers the 8-bit result into p. All 16 flip-flops have an asynchronous, active-low reset. An input applied before edge 1 is captured at edge 1, and the product appears in p at edge 2. The latency is therefore two cycles, and the throughput is one result per cycle.

Fig. 2 shows the gate-level structure that the two `assign` statements describe: two XOR gates form the sum, and three AND gates feeding an OR gate form the majority function for the carry. This is the logic that synthesis later maps onto `xor3` and `maj3` cells.

III. Part A: RTL Design and Verification

A. Testbench

`tb_mult` generates a $10\,\mathrm{ns}$ clock, holds `rst_n` low for the first two cycles and then applies all 256 combinations of a and b. Inputs change on the falling edge. After two rising edges the testbench waits another half cycle and compares p with $a \times b$ (the $\star$ operator is used only here). The comparison uses `!==`, so an unknown output also counts as an error. The testbench prints a line for every mismatch and one summary line, and writes `mult.vcd`. The RTL result is:

```
PASS: 256 vectors checked, 0 errors
```

B. Error-detection check

We temporarily changed the sum in `full_adder.v` to `a ^ b`, which ignores the carry input. The same testbench then reports 95 mismatches, the first being $a = 3$, $b = 3$, $p = 5$ (expected 9):

```
FAIL: 256 vectors checked, 95 errors
```

The change was then undone, and the testbench passes again.

C. Hand-worked example

We used Sudhanva Joshi's roll number (2023102022). Its last digit is 2, so $A = 2 + 5 = 7$, and because $A = 7$ the assignment gives $B = 9$.

$$
\begin{array}{rll}
 & 0\,1\,1\,1 & A = 7 \\
\times & 1\,0\,0\,1 & B = 9 \\
\hline
 & 0\,1\,1\,1 & pp_0\ (b_0 = 1) \\
 & 0\,0\,0\,0 & pp_1\ (b_1 = 0),\ \text{shift } 1 \\
 & 0\,0\,0\,0 & pp_2\ (b_2 = 0),\ \text{shift } 2 \\
 & 0\,1\,1\,1 & pp_3\ (b_3 = 1),\ \text{shift } 3 \\
\hline
 & 0\,0\,1\,1\,1\,1\,1\,1 & P = 63
\end{array}
$$

In the array, $r_0 = 0011$.

- Row 1 adds $pp_1 = 0000$: $s_1 = 0011$, so $p[1] = 1$ and $r_1 = 0001$.

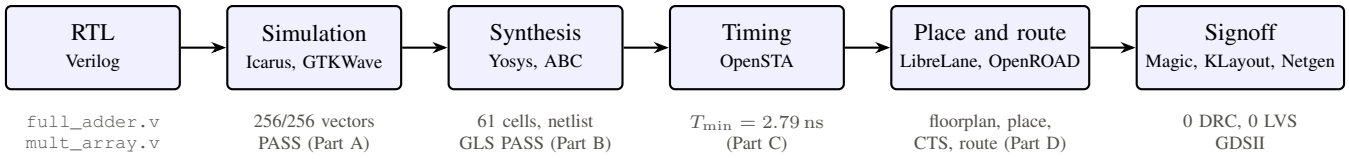


Fig. 1. RTL-to-GDSII flow, the tool used at each stage and the result each stage produced.

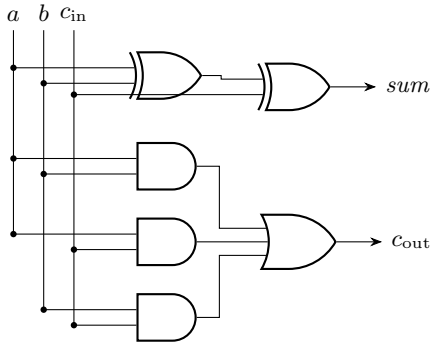


Fig. 2. Gate-level view of `full_adder`: $sum = a \oplus b \oplus c_{\text{in}}$ and $c_{\text{out}} = ab + a\,c_{\text{in}} + b\,c_{\text{in}}$.

- Row 2 adds $pp_2 = 0000$: $s_2 = 0001$, so $p[2] = 1$ and $r_2 = 0000$.
- Row 3 adds $pp_3 = 0111$: $s_3 = 0111$ and $c_3[4] = 0$, so $p[3] = 1$ and $p[7{:}4] = 0011$.

Hence $P = 0011\,1111_2 = 63$.

D. Waveform

Fig. 4 shows the example in the RTL simulation. With $a = 7$ throughout, b changes from 8 to 9 on the falling edge at 2 440 ns. Rising edge 1 at 2 445 ns (marker A) captures a and b into the input registers. Rising edge 2 at 2 455 ns (marker B) captures the product, and p changes from 56 to 63. The latency is two clock cycles, or 20 ns.

IV. Part B: Logic Synthesis

The Yosys script `synth/synth.tcl` performs these steps:

1) read both RTL files;
2) run generic synthesis with the full adders flattened into one module;
3) map the flip-flops, and then the logic (with ABC), to the typical-corner Liberty library;
4) remove unused wires;
5) write the cell and area report and the gate-level netlist.

Table I summarises the results. All cell names omit the library prefix `sky130_fd_sc_hd__`. The flip-flop cell `dfrtp_1` has an area of $25.024\,\mu\text{m}^2$.

We simulated the gate-level netlist against the SKY130 functional cell models with unit gate delays, using the same testbench. It passes all 256 vectors, so synthesis preserved the behaviour. The netlist also contains two `lpflow_isobufsrc_1` cells. ABC is free to pick them because the script does not exclude any library cells.

TABLE I
Part B results

Metric	Value
Total number of cells	61
Number of flip-flops	16
Total cell area	$756.976\,\mu\text{m}^2$
Flip-flop area (16×25.024)	$400.384\,\mu\text{m}^2$
Flip-flop area (share of total)	52.89 %
Three most-used cell types	
`dfrtp_1`	16
`nand2_1`	7
`maj3_1` and `and2_0` (tied)	4 each
Netlist simulation	
`PASS: 256 vectors checked, 0 errors`	

TABLE II
Part C results (typical corner)

Metric	Value
Worst slack at $T = 10$ ns	7.213 ns
$T_{\min}$	2.79 ns
$F_{\max}$	358.4 MHz
Slack at $T = T_{\min}$	0.003 ns
Critical path start point	`_91_` ($a_r[1]$)
Critical path end point	`_88_` ($p[6]$)
Cells on the critical path	8 logic, 2 flip-flops
Latency ($2 \times T_{\min}$)	5.58 ns
Throughput	358.4 million results/s

Fig. 5 groups the 61 cells by function. The 16 flip-flops take more area than all 45 logic cells together. Among the logic cells, the XOR and XNOR cells that form the adder sums are the largest group by area (9 cells, $137.6\,\mu\text{m}^2$): a 3-input XOR is the largest logic cell in the netlist, at $23.8\,\mu\text{m}^2$.

V. Part C: Static Timing Analysis

The OpenSTA script `sta/sta.tcl` reads the typical-corner library and the netlist. It then defines the clock, sets 1.0 ns input and output delays, and reports the worst setup slack and the critical path. At $T = 10$ ns the worst slack is 7.213 ns, so

$$T_{\min} = 10 - 7.213 = 2.787\,\text{ns} \approx 2.79\,\text{ns},$$
$$F_{\max} = 1000/2.79 = 358.4\,\text{MHz}.$$

Rerunning at $T = 2.79$ ns gives 0.003 ns of slack (Table II).

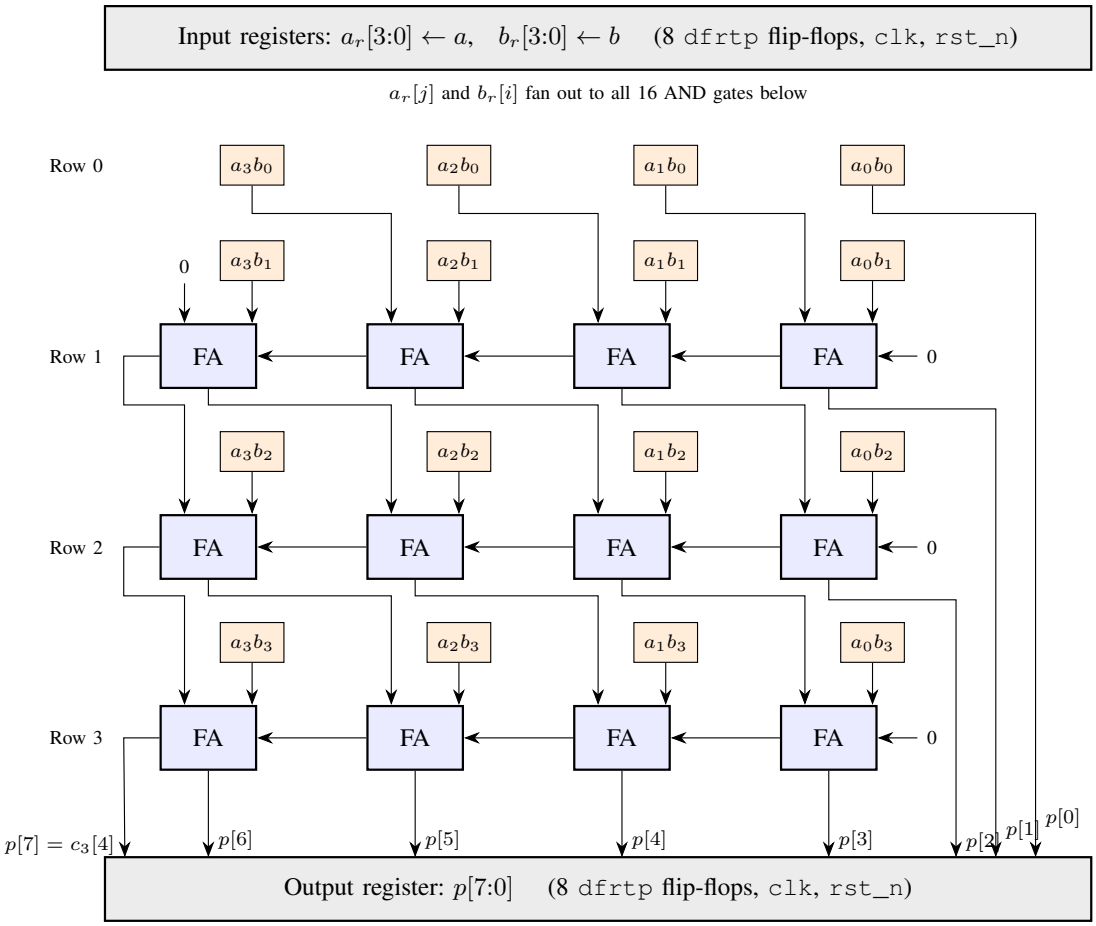


Fig. 3. Block diagram of `mult_array`: 16 AND gates ($a_j b_i = pp_i[j]$), 12 full adders in three ripple-carry rows, and the input and output registers (16 flip-flops). Each full adder takes $pp_i[j]$ on its right input and $r_{i-1}[j]$ on its left input; carries ripple from right to left.

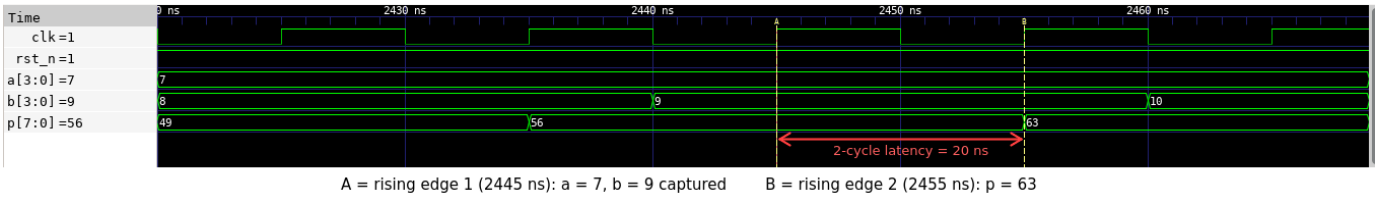


Fig. 4. GTKWave view of `clk`, `rst_n`, a, b and p (decimal) for the example 7×9. Markers A and B are the two rising edges of the 2-cycle latency.

The critical path (Table III) runs from the register holding $a_r[1]$ to the register that captures $p[6]$. Five of its eight logic cells are recognisable full-adder cells: three `maj3` cells (carry) and the `xnor3` and `xor3` cells (sum). The first three cells implement the partial products and the simplified first adder row, which ABC merged into complex gates.

VI. Part D: Physical Design

A. Configuration

Table IV lists the settings in `pnr/config.json`. The clock period is $1.2 \times T_{\min} = 1.2 \times 2.79 = 3.348\,\text{ns}$, rounded up to the next $0.5\,\text{ns}$. As in Part C, timing is signed off at the typical corner only, which the `STA_CORNERS` setting selects.

B. Results

The flow finished in a single run at $3.5\,\text{ns}$ with positive setup slack, 0 DRC errors and 0 LVS errors. Table VII lists the final metrics from `pnr/metrics.json`. The post-route maximum frequency is

$$F_{\max} = \frac{1000}{3.5 - 0.077} = \frac{1000}{3.423} = 292.2\,\text{MHz},$$

$18\,\%$ lower than the $358.4\,\text{MHz}$ from Part C. The post-route critical path runs from the $a_r[0]$ register to the $p[6]$ register through eight logic cells. Four of them are full-adder cells: `xor3`, two `maj3` and `xnor3`. Hold timing is also met, with a worst hold slack of $0.246\,\text{ns}$, and there are no slew, capacitance, fanout or antenna violations.

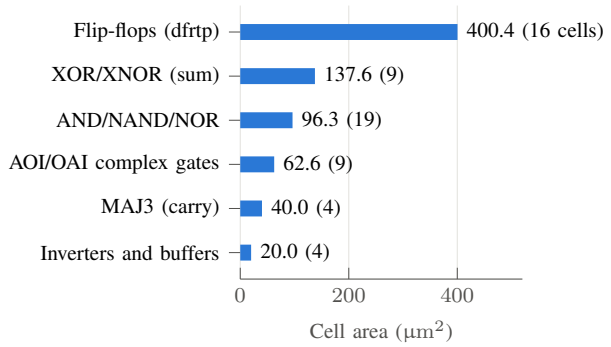


Fig. 5. Synthesised cell area by function (756.976 µm² in total). Labels give the area in µm² and the number of cells.

TABLE III
CRITICAL PATH AT $T = 2.79$ ns

#	Cell	Role	Delay (ns)
–	dfrtp_1 (_91_)	launch, clock to Q	0.398
1	clkinv_1	partial products	0.073
2	o311ai_0	partial products, row 1	0.356
3	a21oi_1	row 1	0.153
4	maj3_1	FA carry	0.402
5	xnor3_1	FA sum	0.370
6	maj3_1	FA carry	0.368
7	maj3_1	FA carry	0.354
8	xor3_1	FA sum	0.178
–	dfrtp_1 (_88_)	capture, setup time	0.135
	Data arrival time		2.652

TABLE IV
LIBRELANE CONFIGURATION

Key	Value
DESIGN_NAME	mult_array
VERILOG_FILES	full_adder.v, mult_array.v
CLOCK_PORT	clk
CLOCK_PERIOD	3.5 ns
FP_SIZING	absolute
DIE_AREA	0, 0, 150, 150 (µm)
STA_CORNERS	nom_tt_025C_1v80

C. Timing after routing

Table V breaks the post-route critical path into its parts. The clock reaches the launch flip-flop 0.324 ns after the clock edge, through two levels of clkbuf_16, and the capture flip-flop 0.290 ns after it. Because the two latencies nearly cancel, the clock tree itself costs little. What costs time is the data path, which takes 3.034 ns from the launch clock pin, against 2.652 ns before layout, and the 0.25 ns clock uncertainty that the flow reserves.

Fig. 6 plots the arrival time at each stage of the two critical paths. Both have a clock-to-Q stage and eight logic stages, but they pass through different cells, so individual stages do not line up. Overall the routed path is 0.38 ns slower. The launch flip-flop now drives seven inputs plus the routed wire, so its clock-to-Q delay rises from 0.398 to 0.476 ns. The rest comes from logic stages driving routed wires, with every delay derated by 5 %.

TABLE V
POST-ROUTE TIMING BUDGET OF THE CRITICAL PATH

Component	Time (ns)
Launch clock latency (clock tree)	0.324
Clock to Q (dfrtp_1)	0.476
Logic and wires (8 cells)	2.558
Data arrival time	**3.358**
Clock period	3.500
Capture clock latency	+0.290
Clock reconvergence pessimism removal	+0.017
Clock uncertainty	−0.250
Setup time (dfrtp_1)	−0.122
Data required time	**3.435**
Setup slack	**0.077**

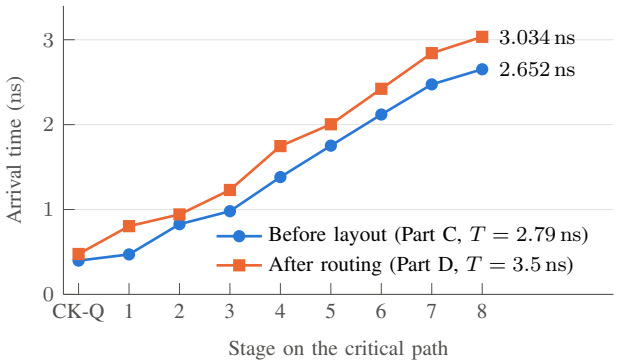


Fig. 6. Arrival time along the critical path before layout and after routing, measured from the launch flip-flop's clock pin. The two paths go through different cells, because LibreLane re-synthesises the design, but both have a clock-to-Q stage and eight logic stages.

D. Power and IR drop

OpenROAD estimates 0.575 mW of total power at the 3.5 ns clock (Table VI), using the flow's default switching activity rather than activity from simulation, so the figure is an estimate. Almost four fifths of it (78 %) is spent in the clock network and the flip-flops: the clock toggles every cycle and drives all 16 flip-flops and the three clock buffers, while the combinational logic switches only when the inputs change. Leakage is negligible at the typical corner (4.7 nW). The worst static IR drop on the power grid is 0.72 mV on VPWR and 0.68 mV on VGND, under 0.05 % of the 1.8 V supply, which is expected for so small a load on a grid sized for the full die.

E. Cells added by the flow

Table VIII breaks down the 1882 instances. LibreLane re-synthesises the RTL itself and produces 62 logic and flip-flop cells. That is one more than Part B, because LibreLane's synthesis does not use the lpflow cells. Every other cell was added during physical design.

TABLE VI
POST-ROUTE POWER AT $T = 3.5$ ns (TYPICAL CORNER)

Group	Internal (µW)	Switching (µW)	Total (µW)	Share
Clock network	149.5	76.3	225.8	39.3 %
Sequential	205.9	16.6	222.5	38.7 %
Combinational	49.4	77.1	126.5	22.0 %
Total	404.8	170.1	574.9	100 %

Leakage: 4.7 nW in total.

TABLE VII
PART D RESULTS (TYPICAL CORNER)

Metric	Value
`CLOCK_PERIOD` used	3.5 ns
Number of runs needed	1
Core area	17 759.5 µm^2
Die area	22 500 µm^2
Cells after synthesis (Part B)	61
Cells after place and route	1882
Kinds of cells added	decap, fill, tap, buffers
Wirelength	2 495 µm
Setup slack	0.077 ns
Post-route $F_{\max}$	292.2 MHz
$F_{\max}$ from Part C	358.4 MHz
DRC errors (Magic, KLayout)	0, 0
LVS errors	0

TABLE VIII
CELLS AFTER PLACE AND ROUTE

Cell kind	Count
Design logic and flip-flops	62
Decap (`decap_3` to `decap_12`)	1300
Fill (`fill_1`, `fill_2`)	253
Well taps (`tapvpwrvgnd_1`)	245
Timing-repair buffers (`buf_2`, `clkdlybuf4s50_1`)	19
Clock-tree buffers (`clkbuf_16`)	3
Total	1882

The die is far larger than the design needs. The cells that do work (logic, flip-flops and buffers) occupy 1 245 µm^2, or 7.0 % of the core. Decap and fill cells cover the remaining 16 515 µm^2 (93 %), so the rows stay continuous for DRC and the spare area adds supply decoupling. Fig. 7 shows where each kind of cell was placed: the logic forms a loose cluster in the centre of the core, the clock and timing-repair buffers sit next to the cells they drive or near the I/O pins, and the critical path climbs from the input register near the bottom to the output register near the top.

F. Layout

Fig. 8 shows the final GDS in KLayout.

- **Whole die:** the 150 µm die, the standard-cell rows and the I/O pins on the die edges. The signal routing is concentrated in the middle of the core, where the 84 logic, flip-flop and buffer cells sit; the rest of the rows hold decap and fill cells.
- **Zoomed view:** the rows bounded by horizontal power rails, the poly gates inside the cells, horizontal metal 1 and vertical metal 2 wires, and the vias between them.

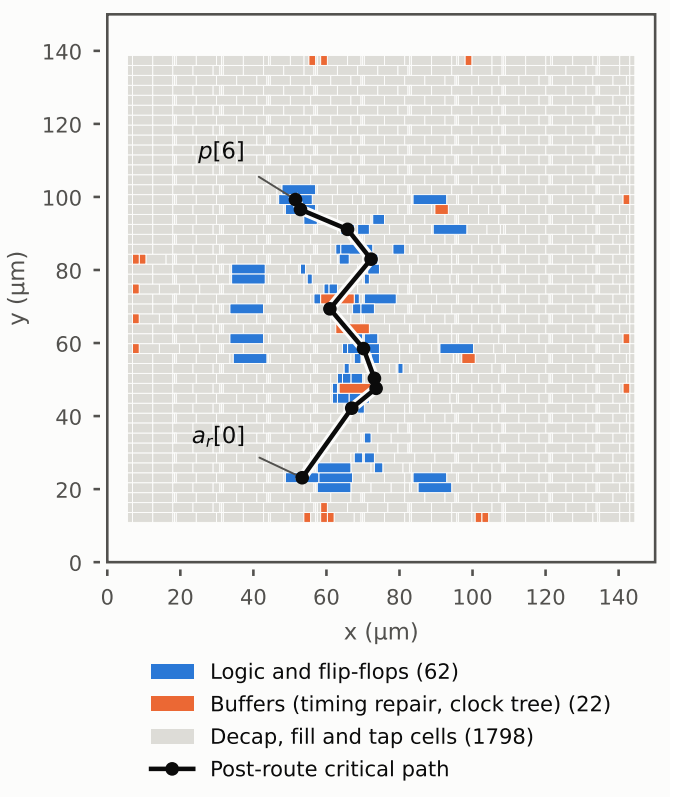


Fig. 7. Placement of all 1882 instances, drawn from the final DEF, with the post-route critical path from the $a_r[0]$ register to the $p[6]$ register overlaid.

G. A note on process corners

By default LibreLane signs off at nine process corners. Before we restricted signoff to the typical corner, exploratory runs with all corners met timing at the typical corner. They failed only at the slow corner (100 °C, 1.60 V), where the path is roughly twice as slow. The README records these runs. The assignment evaluates timing at the typical corner, so the reported run uses that corner only.

VII. ANSWERS TO THE QUESTIONS

Q1. Synthesis produced $61 - 16 = 45$ logic cells for the array, instead of $16 + 12 = 28$ blocks. The count is not one cell per block, because flattening dissolves the full-adder boundaries and ABC re-optimises the whole logic cone. Some effects add cells:

- Some full adders are split into a sum cell (`xor3`, `xnor3` or `xor2`) and a carry cell (`maj3`), or into several smaller gates.
- The library's full-adder cell `fa_1` has two outputs, which ABC cannot map to; it reports it as a multi-output cell

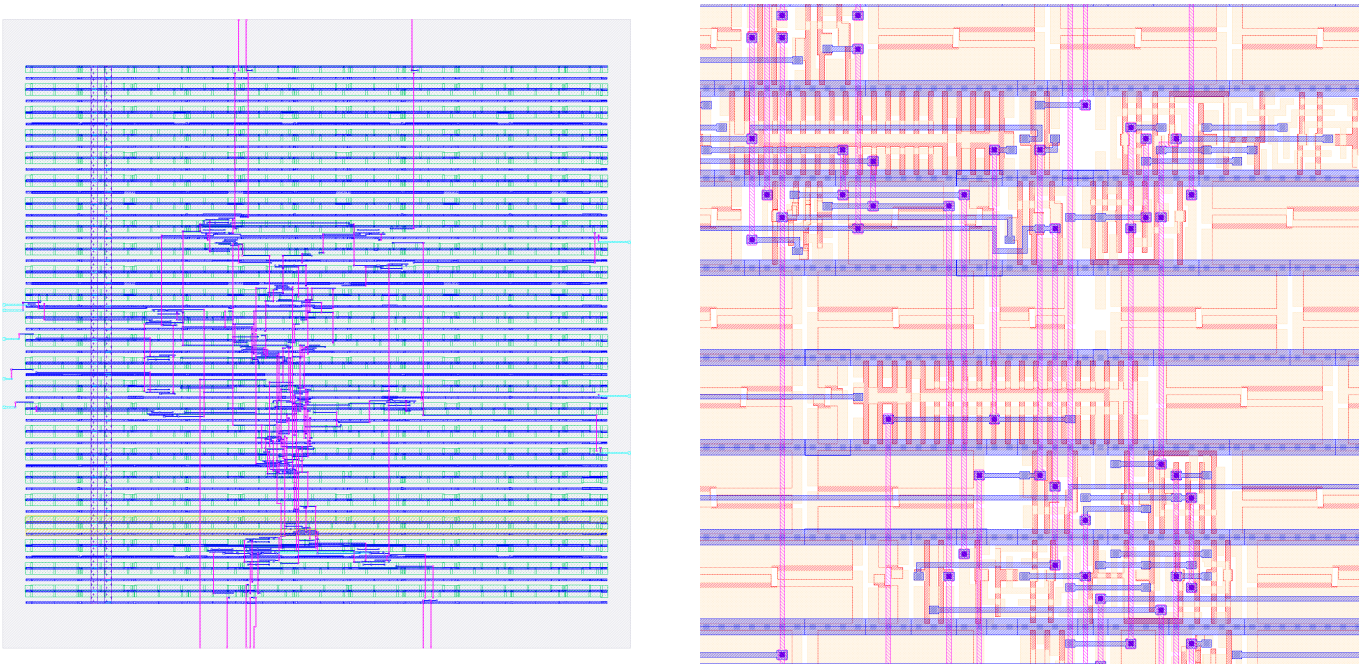


Fig. 8. Final layout in KLayout. Left: whole die (150 µm × 150 µm; metal layers, n-well and boundary). Right: zoomed view of about 20 µm × 20 µm (poly, local interconnect, metal 1, metal 2 and vias).

and skips it. Each full adder therefore costs at least two cells.

Others remove cells:

- Adders with a constant-0 input ($c_i[0] = 0$, $r_0[3] = 0$) reduce to half adders.
- AND gates are merged into complex cells such as `a21oi`, `a22oi` and `o311ai`.

Q2. At $T_{\min} = 2.79\,\text{ns}$, the critical path runs from the input-register flip-flop holding $a_r[1]$ to the output-register flip-flop that captures $p[6]$. It has eight logic cells between the two flip-flops (Table III), five of them full-adder cells. The estimate of $2N = 8$ full-adder delays describes the unoptimised array. Here the first stages are simplified into three fast complex gates (half adders and merged AND logic), so only five full-adder stages remain. Those five cells take $1.672\,\text{ns}$, about $0.33\,\text{ns}$ each, and the three first-stage gates take $0.582\,\text{ns}$. The total logic delay of $2.254\,\text{ns}$ therefore corresponds to about seven full-adder delays, slightly below the estimate of eight.

Q3. Flip-flops take $52.89\,\%$ of the total cell area: 400.384 of $756.976\,\mu\text{m}^2$. A `dfrtp_1` with asynchronous reset is $25.0\,\mu\text{m}^2$, about seven times a `nand2_1` ($3.75\,\mu\text{m}^2$), because it contains two latches, a clock inverter and reset logic. The design needs 16 of them (8 input bits and 8 output bits), while the combinational array is only 45 small gates. The number of registers grows linearly with N, but the array grows with N^2, so the flip-flop share is largest for small multipliers.

Q4. The post-route $F_{\max}$ is $292.2\,\text{MHz}$, against $358.4\,\text{MHz}$ in Part C. Three effects lower it:

- The routed wires add resistance and capacitance, which are extracted after routing.
- The flow adds a $0.25\,\text{ns}$ clock uncertainty and a $5\,\%$ timing derate.
- The inputs are driven by a real cell and the outputs drive a real load.

Together these stretch the launch-to-capture data path from $2.652\,\text{ns}$ to $3.03\,\text{ns}$. The flow added five kinds of cells (Table VIII), each for a reason:

- **Clock-tree buffers** (`clkbuf_16`) deliver the clock to all 16 flip-flops with low skew. They add $0.32\,\text{ns}$ of insertion delay, which is common to the launch and capture flip-flops.
- **Timing-repair buffers** fix slew and capacitance limits and improve timing.
- **Tap cells** tie the n-wells and the substrate to VPWR and VGND, preventing latch-up.
- **Decap cells** add capacitance between the supply rails, reducing supply noise when cells switch.
- **Fill cells** close the remaining gaps in the rows, so that the wells, implants and power rails are continuous, as the design rules require.

Q5. A DRC (design rule check) verifies that the layout geometry obeys the foundry's manufacturing rules: minimum widths, spacings, enclosures, overlaps and densities on every layer. An LVS (layout versus schematic) check extracts the transistors and their connections from the layout and compares them with the netlist. It confirms that the layout implements exactly the intended circuit, with no shorts, opens or missing devices. Both must pass before fabrication:

- A DRC violation can make the chip impossible to manufacture reliably, for example through shorts, broken wires

or low yield.

- An LVS mismatch gives a chip that can be manufactured but does the wrong thing.

Our final layout has 0 DRC errors (Magic and KLayout) and 0 LVS errors (Netgen).

Q6. For an 8-bit multiplier ($N = 8$):

- **Size:** the array needs $N^2 = 64$ AND gates and $N(N-1) = 56$ full adders, against 16 and 12 for $N = 4$. That is about 4.7 times as many full adders, so the logic area grows roughly with N^2.
- **Path length:** the critical path is about $2N = 16$ full-adder delays instead of 8, twice as long. The delay grows linearly with N.
- **Delay estimate:** in Part C, the 4-bit logic delay is $2.652 - 0.398 = 2.254\,\mathrm{ns}$. Scaling it by $16/8$ gives about $4.5\,\mathrm{ns}$ of logic. Adding the clock-to-Q time ($0.40\,\mathrm{ns}$) and the setup time ($0.14\,\mathrm{ns}$) gives $T_{\min} \approx 5.0\,\mathrm{ns}$, or $F_{\max} \approx 200\,\mathrm{MHz}$, against $358\,\mathrm{MHz}$ for the 4-bit design.
- **Registers:** the design would need $8 + 8 + 16 = 32$ flip-flops.

VIII. Conclusion

We took the 4-bit array multiplier from RTL to a DRC- and LVS-clean SKY130 layout. It is functionally correct in both RTL and gate-level simulation. At the typical corner it runs at $358\,\mathrm{MHz}$ before layout and $292\,\mathrm{MHz}$ after routing, on a $150\,\mu\mathrm{m} \times 150\,\mu\mathrm{m}$ die. All source files, scripts and generated reports are in the repository at https://github.com/SudhuJ/arrayMultiplier, whose README lists every command, in order.